

AI FUNDAMENTALS

मिति: २०८३ असोज ८ गते
समय: बिहान ११:०० बजे
स्थान: NHEICC सभाहल, टेकु, काठमान्डौ।



नेपाल सरकार
स्वास्थ्य तथा खाद्य स्वच्छता मन्त्रालय
राष्ट्रिय स्वास्थ्य शिक्षा, सूचना तथा सञ्चार केन्द्र



Mahesh Dahal

LinkedIn

[linkedin.com/in/maheshdahal](https://www.linkedin.com/in/maheshdahal)

Facebook

[facebook.com/maheshdahal](https://www.facebook.com/maheshdahal)



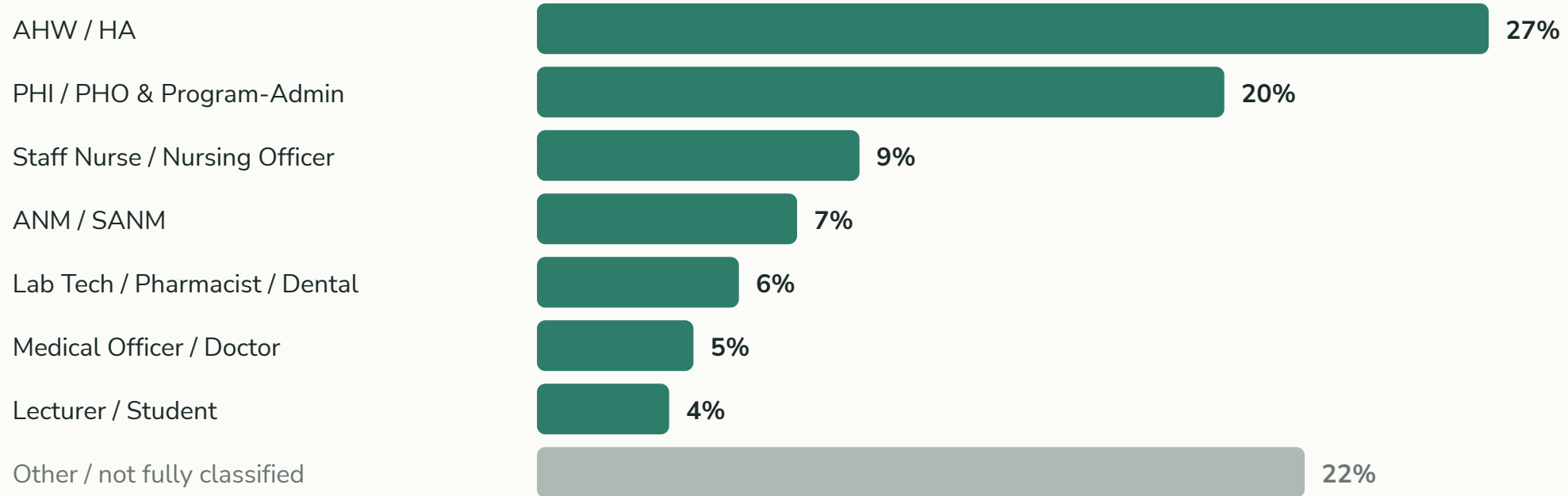
नेपाल सरकार
स्वास्थ्य तथा खाद्य स्वच्छता मन्त्रालय
राष्ट्रिय स्वास्थ्य शिक्षा, सूचना तथा सञ्चार केन्द्र

Expectations

- 2 Hours Session
- Zoom Link and Facebook
- Demo led session
- Content is set to Average Audience
- Please put the questions on comment and chat
- **Not in today's Session:**
- **Model Tuning/building, vector DB, Tokenization, Cosine Similarity**

WHO'S IN THE ROOM

~7500+ registrants, nationwide



Public Health as common thread

Main Agenda

- एआईका आधारभूत कुराहरु
- वेबसाइट, इन्फोग्राफिक्स र भिडियो
- RAG र NotebookLM
- llama र Qwen
- के गर्ने । के नगर्ने

एआईका आधारभूत कुराहरु

What Is AI, in Plain Terms

Think of it as a very well-read assistant who has read an enormous number of documents and can write, summarize, or explain things in seconds — but who sometimes gets details wrong, and has never met your patients or your community.

Good at

Drafting, rephrasing, translating, summarizing long documents, answering general questions, and creating images, audio, or video from a description.

Not

A verified-fact search engine, a doctor, or a replacement for judgement — and it doesn't know your audiences or local guidelines unless you tell it.

एआईका आधाररभूत कुराहरु

AI Output Is Never Final

Everything AI produces is a first draft. It always needs post-processing — editing, correction, and verification — before it's ready to use.

Video

Needs further editing. AI does not replace traditional video tools.

Software

Needs testing and debugging. AI does not replace engineers.

Flyer / Document

Needs correction and editing before it's ready to share.

The work is always a draft. Always assume it has mistakes.

How Is AI "Trained"?

Before you ever type a prompt, the model was shown enormous amounts of text and adjusted, again and again, until it got good at predicting language patterns. All of that learning already happened — typing to it doesn't teach it anything new about the world.

Think of it like...

A health worker's years of study and clinical practice before their first patient — by the time you ask it a question, the "training" is long finished.

[HTTPS://TEACHABLEMACHINE.WITHGOOGLE.COM/](https://teachablemachine.withgoogle.com/)



Give Role + Task + Audience + Format in every prompt



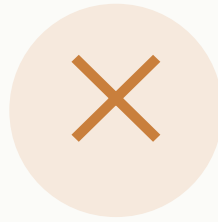
**Post-process and verify before publishing
anything**



**Use Simple Language, Specially for
नेपाली**

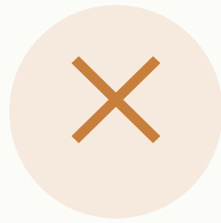


Verify facts, figures, and details yourself

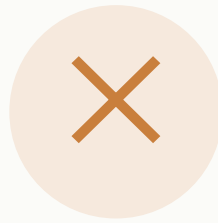


Never enter PII* into a public AI tool

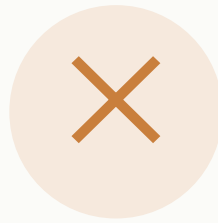
*Personally-Identifiable Information



**Don't trust AI-generated content without expert
review**



**Don't use AI tools from
sketchy vendors**



**Don't assume the first draft is
ready to use as-is**

Conversational Gemini on Your Phone

gemini.google.com, or the Gemini app — often pre-installed on Android. Free, no payment needed, sign in with a Google account. Use it like a knowledgeable assistant for everyday questions.

The prompt formula

Role + Task + Audience + Format. Example: "You are a health educator. Write a short message about hand hygiene for mothers, in simple Nepali, under 100 words."

Beyond Chat: Video, Audio, Images & Text

Gemini and Google AI Studio (aistudio.google.com) can generate images and help draft text, all from one plain-language description — the same conversational style you just used.

Coming up next

Dedicated tools go deeper on each of these, one at a time: common mistakes to avoid, then Suno for songs and jingles, HeyGen & Flow for talking-head video, and AI Studio for building small websites and software.

THEORY

Common Mistakes & How to Fix Them

Prompting mistakes — vague requests, trusting the first draft — apply everywhere. Here's what breaks specifically in images, video, and audio.

Gemini Images — Nepali Script Gets Scrambled

Mistake: Asking it to put Nepali/Devanagari text inside a generated image — it almost always comes out garbled (see the example on the next slide).

Pro tip: Generate the image with no text, then add the Nepali caption yourself afterward in an editor. Don't trust AI to render script.

Video — The Voice Doesn't Sound Right

Mistake: The generated voice sounds flat, mispronounces Nepali words, or has the wrong tone for the message.

Pro tip: Regenerate — the voice varies each run. If the tool lets you pick a voice or language setting, choose it explicitly, and always listen to the whole clip before using it.

Audio — Suno

A generic style prompt gives a random genre, and lyrics written like a sentence sound flat. Name the genre, instrument, mood, and length exactly, and write lyrics the way they're meant to be sung.

Video — HeyGen & Flow Rendering

A script written like a document makes the avatar sound robotic — read it aloud first. Rendering takes 1–3 minutes: reloading mid-render cancels the job.

Video & Images — Fingers, Faces & Outfits Keep Changing

Mistake: Hands come out with too many or oddly bent fingers, and a person's face or outfit can shift between shots — AI struggles to stay consistent across a video or a series of images.

Pro tip: Avoid close-ups on hands, generate a few takes and pick the cleanest one, and keep to one continuous generation rather than stitching separate clips together when the look needs to match.

EXAMPLE · GEMINI IMAGE GENERATION



This is what happens when you ask for Devanagari text inside a generated image — garbled, unreadable script.

Audio for Community Messaging: Suno

Suno (suno.com) generates full songs or jingles from a text description — useful for radio spots, loudspeaker announcements, or a WhatsApp voice note. Most directly useful for PHI/PHO and program/IEC staff.

Ready-made jingle — "आमाको दूध, बच्चाको हक" (Mother's Milk, the Child's Right)

आमाको दूध, बच्चाको हक — पहिलो घण्टैदेखि, छ महिना एक!

Gloss: Mother's milk is the child's right — from the first hour, exclusive for six months. Built on Nepal's two core breastfeeding messages.

Style prompt: "Warm, gentle Nepali folk-pop lullaby, female vocal, soft acoustic guitar with light hand percussion (madal), tender mood, ~25–30 seconds."

Video for Awareness Campaigns

HeyGen

Turns a script into a video of an AI presenter speaking it, with lip-synced avatars. Good for a talking-head health message without filming anyone. Rendering takes 1–3 minutes — start it first, come back later.

Google Flow

Generates short video clips from a text description or a still image, built on Google's Veo model. Access often needs a paid Google AI plan — confirm before planning a live demo.

Frame both the same way: AI gives a fast first draft, not a final product — review before anything goes out publicly. Most directly useful for PHI/PHO and program/IEC staff; others can treat it as an awareness demo.

AI Studio — Building Software & Websites

Google AI Studio (aistudio.google.com) can write working software from a plain-language description — no coding experience needed. This is sometimes called "vibe coding": you describe what you want, AI writes the code, and you see it running right away.

Try asking

- "Build a simple webpage for our health post with our name, address, opening hours, and services, in Nepali."
- "Build a small calculator: enter a child's date of birth, it tells me which vaccines are due this month."
- "Build a one-page form to record daily patient counts, with a button that totals them."

Fast to build, still needs review — the "always a draft" rule applies to code too.

Which Model is Best?

LiveBench (livebench.ai) is a public leaderboard that tests AI models on fresh questions every month — so a model can't just memorize the answers in advance. It's one honest way to compare tools, including open-weight models like Llama and Qwen, which we just covered.

What it tests

Math, coding, reasoning, language, instruction-following, and data analysis — scored automatically, not guessed by another AI or a human's opinion.

<https://livebench.ai/#/>

THEORY

Challenges with AI

Before we go further, four honest limits worth knowing — the last one is what today's next topic actually fixes.

1. Cost

Free tiers run out fast, and paid subscriptions add up — especially if MoHP or NHEICC wants everyone using this at scale.

2. PII & Privacy

Every prompt sent to a public tool leaves your network. Patient-identifiable information should never go into it.

3. Hallucination

It can sound completely confident and still be wrong — dates, dosages, and figures need checking every time.

4. Doesn't Have Your Knowledge

It hasn't read Nepal's guidelines, your HMIS reports, or this month's data — unless you give them to it. That's next.

Meet Llama & Qwen

Open-weight AI models you can download and run yourself — instead of only through a company's website or app.

Llama

Built by Meta. One of the most widely used open-weight model families — free to download, in different sizes to fit different hardware.

Qwen

Built by Alibaba. Free to download and run, with strong performance across many languages beyond English.

Closed tools — ChatGPT, Gemini, Claude

Run on the company's own servers. Every prompt and document you send leaves your network.

Open-weight tools — Llama, Qwen

Run on your own hardware. Nothing has to leave your building or your country.

THEORY

Why This Matters for Nepal's Health System

Not for today's demos — but worth knowing this option exists.

Patient Data Stays Put

Runs offline, on your own machine or server. Nothing about a patient ever has to leave the health post or the ministry's network.

Works Without Reliable Internet

No connection needed once it's set up — useful for health posts in remote districts with weak or no internet.

No Recurring Cost

Free to run once installed. No monthly subscription — useful if MoHP or NHEICC wants to deploy this at scale.

Can Be Built On Nepal's Own Data

MoHP or NHEICC could adapt it using Nepal's own guidelines and language, without sending anything to an outside company.

What Is RAG?

Retrieval-Augmented Generation: instead of answering only from what it learned during training, the AI first reads documents you give it — then answers using those documents.

Think of it like...

Handing a new health worker the actual district protocol book before they answer a question — instead of relying only on what they memorized back in training.

Regular AI

Answers from general training only. May not know Nepal's specific guidelines, forms, or this month's figures.

AI + RAG

Answers using the exact document you upload — a guideline, circular, or report — and can quote it directly.

Where RAG Actually Helps

MoHP Treatment Protocols

Upload the current protocol PDF. Ask: "What's the standard dosage for this drug in a child under 5?" The answer comes straight from the protocol, not general knowledge.

NHEICC IEC Materials

Upload NHEICC's already-approved health messages. Ask: "Does this new flyer draft match our approved messaging on breastfeeding?" Keeps new material consistent with what's signed off.

HMIS Reporting Manual

Upload the HMIS manual. Ask: "Which indicators go in Section 4, and how are they defined?" No more hunting through a hundred-page PDF for one definition.

District & Annual Reports

Upload last year's district health report. Ask: "Summarize the immunization trends from this report in 3 bullet points." A long document becomes usable in minutes.

<https://Notebooklm.google>

DEMO

NotebookLM — Try It Live

notebooklm.google is Google's free RAG tool — upload any document and it answers only from what you gave it, with the source shown right alongside the answer.

Live now

- Upload one real document from the room — a protocol PDF, an HMIS manual, or a district report.
- Ask it a direct question about that document.
- Show how the answer points back to the exact page or line it came from.

<https://notebooklm.google>

Thank You

NHEICC AI Training for Health Workers

Questions after today? Reach out through your usual NHEICC contact point.